A Progressive Registration-Fusion Co-Optimization A-Mamba Network: Toward Deep Unregistered Hyperspectral and Multispectral Fusion

Zan Li, *Fellow, IEEE*, Yue Wen, Song Xiao, *Member, IEEE*, Jiahui Qu, *Member, IEEE*, Nan Li, and Wenqian Dong, *Member, IEEE*

***Abstract*— The existing methods of hyperspectral image (HSI) and multispectral image (MSI) fusion usually overlook the fact that multisource images acquired under different imaging conditions are generally not perfectly registered. Despite the many such methods that have begun to address registration issues, it is still a challenge that most works perform registration and fusion as two separate steps, resulting in a cumulative error. To address this challenge, we propose a progressive registration-fusion co-optimization A-Mamba network (PRFCoAM), which iteratively optimizes the modal-aligned progressively registration-fusion (MAPRF) module to adaptively corrects the deformation from an extensive to a detailed level and refines the fusion results at each level to achieve progressive registration-fusion co-optimization. The proposed MAPRF module integrates the modal-unified local-aware registration (MULAR) block and the interactive attention Mamba fusion (IAMF) block, which facilitates the network comprehensively and efficiently capture features of different levels. Specifically, MULAR adaptively learns spectral and spatial degradation functions to transform the input images into a unified modality and progressively repairs nonrigid pixel offsets by capturing the correlations and differences between corresponding regions of images. IAMF multidirectionally scans the spatial and spectral global dependent features of the well-registered images, which can stimulate the potential of Mamba in fusion and achieve a win–win situation of computational efficiency and selectivity advantages in the global acceptance domain. Extensive experiments demonstrate that PRFCoAM can flexibly deal with different degrees and kinds of nonrigid deformation and achieves state-of-the-art performance. The code will be available at https://github.com/Jiahuiqu/PRFCoAM-for-HSI-MSI-Registration-Fusion**

***Index Terms*— Co-optimization, hyperspectral image (HSI), Mamba, unregistered image fusion.**

Received 16 March 2025; revised 1 May 2025; accepted 12 May 2025. Date of publication 16 May 2025; date of current version 29 May 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62201423, Grant 62471359, and Grant 62476013; in part by the Key Research and Development Program of Shaanxi under Grant 2025SF-YBXM-513; in part by the Young Talent Fund of Association for Science and Technology in Shaanxi under Grant 20230117 and Grant 20250133; in part by the Young Talent Fund of Xi'an Association for Science and Technology under Grant 959202313052; and in part by the Fundamental Research Funds for the Central Universities under Grant QTZX25084. *(Corresponding authors: Jiahui Qu; Song Xiao.)*

Zan Li, Yue Wen, Jiahui Qu, and Wenqian Dong are with the State Key Laboratory of Integrated Service Network, Xidian University, Xi'an 710071, China (e-mail: zanli@xidian.edu.cn; 23011210456@stu.xidian.edu.cn; jhqu@xidian.edu.cn; wqdong@xidian.edu.cn).

Song Xiao is with the Department of Electronic and Communication Engineering, Beijing Electronic Science and Technology Institute, Beijing 100070, China, and also with the School of Telecommunications Engineering, Xidian University, Xi'an 710071, China (e-mail: xs_xidian@163.com).

Nan Li is with Anhui Province Key Laboratory of Physical Geographic Environment, Anhui Engineering Research Center of Remote Sensing and Geoinformatics, Anhui Center for Collaborative Innovation in Geographical Information Integration and Application, Chuzhou University, Chuzhou 239000, China (e-mail: linan@chzu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3570954

I. Introduction

HYPERSPECTRAL images (HSIs) contain hundreds of continuous spectral bands and can reflect fine discriminative features for different ground object recognition [1]. However, long exposures are essential for hyperspectral systems to achieve a sufficient signal-to-noise ratio (SNR), which results in low spatial resolution of HSIs [2], [3]. The low spatial resolution limits the application of HSIs in various remote sensing tasks, including image classification [4], [5], target detection [6], [7], and anomaly detection [8]. By comparison, multispectral systems can acquire multispectral image (MSI) with high spatial resolution (HR MSI). Naturally, a cost-effective way of improving the spatial resolution of HSIs is to combine low-spatial-resolution HSI (LR HSI) with HR MSI of the same scene. This process, known as HSI/MSI fusion, combines the advantages of both HSIs and MSIs to produce images with high spatial and spectral resolution [9], [10].

The HSI/MSI fusion methods are typically categorized into two types: traditional methods and deep learning (DL) methods. Traditional methods include component substitution (CS), multiresolution analysis (MRA) methods, Bayesian methods, and matrix factorization. The CS methods involve decomposing the HSI, substituting its spatial information with the high-resolution image and ultimately constructing the fused HSI by the corresponding inverse transformation. Aiazzi et al. [11] built a general scheme considering the impact of spectral response function in the CS procedure. The MRA methods extract the high-frequency component obtained through the multiresolution decomposition of the panchromatic image (PAN) and incorporate it into each band of the LR HSI. Although these methods benefit from speed and simplicity and excel in preserving spatial information, they often exhibit significant spectral distortion [12].

Bayesian methods typically use appropriate prior distributions for input HR MSIs and LR HSIs to solve an optimized model. Akhtar et al. [13] first proposed a Bayesian

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:13:02 UTC from IEEE Xplore. Restrictions apply.

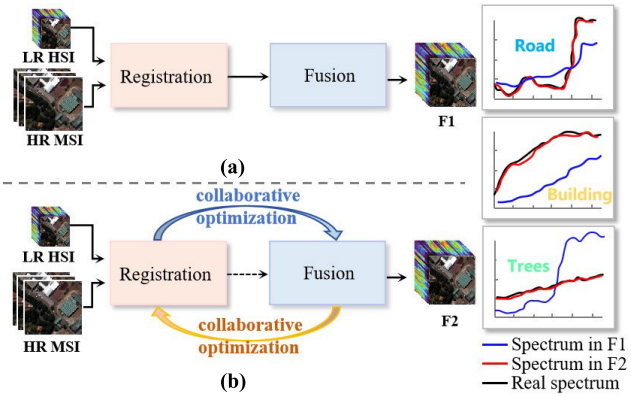


Fig. 1. Analysis of the importance of registration-fusion co-optimization to HSI fusion results. (a) Registration and fusion are carried out independently. (b) Registration and fusion are carried out cooperatively.

sparse representation-based approach. Matrix decomposition approaches assume that the spectrum of each pixel can be estimated as a combination of multiple spectral signature bases. However, these handcrafted priors, often simplistic and subjective, fail to adequately capture the inherent complexity of spatial and spectral configurations in real HSIs.

DL methods learn the appropriate parameters based on the supervision of training samples [14], [15], which can reduce the dependence on prior knowledge and gradually become the mainstream method of image fusion. Wu et al. [16] obtained the desired HR HSI by transforming the HSI/MSI fusion into spatial SR (SpaSR) and spectral SR (SpeSR) modules that constrain each other. Fu et al. [17] proposed MixFus, a mixed noise-oriented fusion method combining sparse noise detection, subspace enhancement, and regularization for robust HSI–MSI fusion without prior training. Liu et al. [18] proposed a cyclic network (Circle-Net) that generates fused images from only one pair of source images without prior training of network parameters. In [19], the multihierarchical cross Transformer (MCT) was developed to facilitate cross-modality information interaction between features at identical scales. Most current DL methods are designed based on the premise that the two images are strictly registered. However, since the perceptual disparities between images captured by different sensors and the transformation of images can be nonlinear distortion, it is challenging to achieve precise registration. If the images are not correctly registered, the same target in different images may misalign, resulting in the fused image exhibiting blurring, ghosting, or other artifacts.

Recently, some works have focused on the unregistration problem in the fusion of HSI/MSI [20]. Ying et al. [21] introduced the normalized edge difference (NED) to align images by minimizing edge discrepancies. Zhang et al. [22] proposed HMF-DicL to reduce spectral mismatches caused by misregistration in traditional HMF methods. However, these methods carry out the registration and fusion tasks separately, where the fusion results fail to provide feedback to improve the previous registration accuracy and lead errors accumulate. Fig. 1 shows the framework of two-stage registration-fusion (a) and registration-fusion co-optimization (b), respectively. The spectral curves of various objects in the fused image obtained by the network of Fig. 1(b) are closer to the real spectrum. Zheng et al. [23] presented a nonrigid registration HSI-SR network (NonRegSRNet) that integrates the registration and fusion tasks into a unified model. NonRegSRNet renders the two tasks mutually complementary, but relying solely on local convolution kernels limits the ability to capture long-range dependencies in HSIs and MSIs due to restricted receptive fields. Different from the NonRegSRNet, Guo et al. [24] proposed a stereoscopic cross-attention network (SCANet) based on a Transformer that integrates registration and fusion tasks into a unified model. Compared to the CNN, the Transformer better captures long-range dependencies, enhancing global feature representation. However, the self-attention mechanism introduces a computational bottleneck, as its complexity scales quadratically with input size, leading to substantial overhead in high-resolution tasks. To mitigate this, researchers have improved the efficiency of attention by constraining the size or stride of computing windows [25], [26]. While effective, such methods force a compromise between effective receptive field and computational cost, hindering long-range dependency modeling. It is therefore necessary to develop a new architecture for efficient visual data modeling while preserving the global receptive fields feature of self-attention.

The state-space model (SSM) has become a central focus in recent developments [27]. Compared to the Transformer, the SSM addresses the challenge of establishing long-range dependencies by carrying out state transitions and implementing them through convolutional operations, which results in near-linear computational complexity [28]. Mamba further enhances the computational efficiency of the SSM mainly through two key innovations [29]: hardware optimization and the utilization of time-varying parameters. Time-varying parameters mean that it features an input-dependent mechanism, dynamically adjusts SSM parameters, and enhances information filtering. Hardware optimization means that Mamba processes data linearly with sequence length, enhancing computational speed in modern systems. Unlike the Transformer, which relies on quadratic-complexity attention mechanisms, Mamba is a purely SSM-based model that enables each element in a 1-D sequence (e.g., a text sequence) to access contextual knowledge exclusively through a compressed hidden state. This design which effectively compresses the global context with an input-dependent selective SSM reduces the computational complexity from quadratic to linear [27], [30] while maintaining the benefits of global receptive fields. By providing a selective dependency mechanism, efficient GPU memory usage, and linear scalability with context length, Mamba bridges the gap between the traditional SSM and contemporary neural network frameworks and provides an effective solution for a wide range of sequential data-processing tasks [29].

A series of modules based on Mamba [31], [32] were introduced in many computer vision tasks, including image classification and semantic segmentation. We note the excellent computational efficiency of Mamba in natural image processing and want to excavate development potential in HSI/MSI fusion. However, there is still a problem that needs

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:13:02 UTC from IEEE Xplore. Restrictions apply.

to be solved: due to the unidirectional information induction attribute of the Mamba, the feature extraction mechanism in a single direction may not fully capture the context information between spectral bands of HSIs and understand the spatial correlation of MSIs.

In conclusion, HSI fusion with unregistered image tasks still has two major challenges.

1) Carrying out the fusion and registration tasks independently means that the fusion results rely on the accuracy of the previous registration, further leading to the accumulation of errors. Could we design a framework to guide the interaction and joint optimization of registration and fusion?
2) The development of Mamba offers a promising path for balancing global receptive fields with computational efficiency. How can we further explore the potential of Mamba in remote sensing image fusion?

To solve the above problems, we propose a progressive registration-fusion co-optimization A-Mamba network (PRFCoAM) consisting of a modal-aligned progressively registration-fusion (MAPRF) module. By progressively performing MAPRF from extensive to detailed levels, PRFCoAM adaptively corrects deformation at each level and progressively optimizes the fusion of well-registered images to achieve registration-fusion co-optimization. MAPRF integrates the modal-unified local-aware registration (MULAR) block and the interactive attention Mamba fusion (IAMF) block. MULAR learns the meaningful spectral response function and the point spread function (PSF) through the constraint of the loss function in the physical observation model, thus transforming the input image under a unified modality to assist fix pixel offset in the same reference standard. Under the unified modality, MULAR adaptively matches the most accurate pixel points and corrects the spatial deformation in the small neighborhood by embedding local attention to obtain a well-registered image. IAMF implants the attention and multidirectionally scans the spatial and spectral features of input images, which overcomes the limitations of extracting continuous spectral and spatial features caused by the unidirectional information induction of Mamba and explores the potential of Mamba in fusion. The main contributions of this article are as follows.

1) We propose a PRFCoAM, which optimizes the results of registration and fusion from an extensive to detailed scale in an iterative framework to obtain the desired HR-HSI.
2) We propose the MULAR block, which adaptively learns the meaningful spectral and spatial degradation functions to transform the input image into a unified modality and embed local attention that focuses on the correlations and differences between corresponding regions of images to fix the nonrigid pixel offset.
3) We propose an IAMF block, which multidirectional scans feature to overcome the dilemma of extracting continuous spectral and spatial features caused by Mamba and implanting the attention to grasp the most crucial information that contributes to fusion.

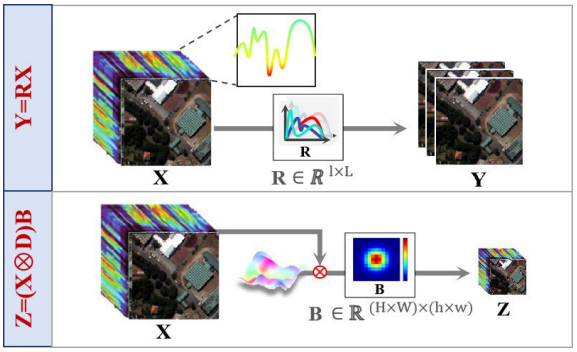


Fig. 2. Schematic of the $\boldsymbol{R}$ and $\boldsymbol{B}$ mechanism.

The rest of the article is organized as follows. Section II describes the proposed PRFCoAM in detail. Section III presents and analyzes the experimental results on four datasets. The conclusion is provided in Section IV.

II. Proposed Approach

A. Problem Formulation

In this article, we focus on how to achieve HR HSI fusion using HSIs and MSIs that have not been precisely registered. We assume that the HR MSI acts as the fixed image $\mathbf{Y} \in \mathbb{R}^{H\times W\times l}$, as it comprises detailed spatial information of ground objects, while the LR HSI acts as the moving image $\mathbf{Z} \in \mathbb{R}^{h\times w\times L}$. We aim to obtain the ideal HR HSI $\mathbf{X} \in \mathbb{R}^{H\times W\times L}$ from $\mathbf{Y}$ and $\mathbf{Z}$, where $\{H, h\}$, $\{W, w\}$, and $\{L, l\}$ denote the length, width, and the number of spectral images, respectively ($l \ll L$, $h \ll H$, and $w \ll W$). According to Wald's protocol, $\mathbf{Y}$ and $\mathbf{Z}$ can be regarded as degraded images of $\mathbf{X}$ in the spectral and spatial dimensions, respectively. In addition, due to different imaging conditions, the acquired $\mathbf{Z}$ is often geometrically distorted and may not maintain a complete spatial correspondence with $\mathbf{Y}$. The relationship between them can be expressed as

$$\mathbf{Y} = \mathbf{RX} \tag{1}$$

$$\mathbf{Z} = (\mathbf{X} \otimes \mathbf{D})\boldsymbol{B} \tag{2}$$

where $\boldsymbol{R} \in \mathbb{R}^{l\times L}$ and $\boldsymbol{B} \in \mathbb{R}^{(H\times W)\times(h\times w)}$ denote the spectral response function (SRF) and PSF, and $\mathbf{X} \otimes \mathbf{D}$ represents the deformation field $\mathbf{D}$ acts on $\mathbf{X}$. As shown in Fig. 2, (1) models the transformation relationship between the spectral resolution of $\mathbf{X}$ and $\mathbf{Z}$ and (2) models the spatial–spectral relationship between $\mathbf{X}$ and $\mathbf{Y}$ (i.e., the relationship between spatial resolution and the local spectral inconsistencies caused by deformation between $\mathbf{X}$ and $\mathbf{Z}$).

Existing DL methods for HSI/MSI fusion based on unregistered images commonly handle registration and fusion as separate processes

$$\begin{cases} \mathbf{Z}_{\text{cor}} = \mathcal{D}(\mathbf{Y}, \mathbf{Z}) \\ \mathbf{X} = \mathcal{F}(\mathbf{Z}_{\text{cor}}, \mathbf{Y}) \end{cases} \tag{3}$$

where $\mathcal{D}(\cdot)$ represents the image registration network, $\mathcal{F}$ represents the HSI fusion network, and $\mathbf{Z}_{\text{cor}}$ is the registered LR HSI. These methods separate registration and fusion into

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:13:02 UTC from IEEE Xplore. Restrictions apply.

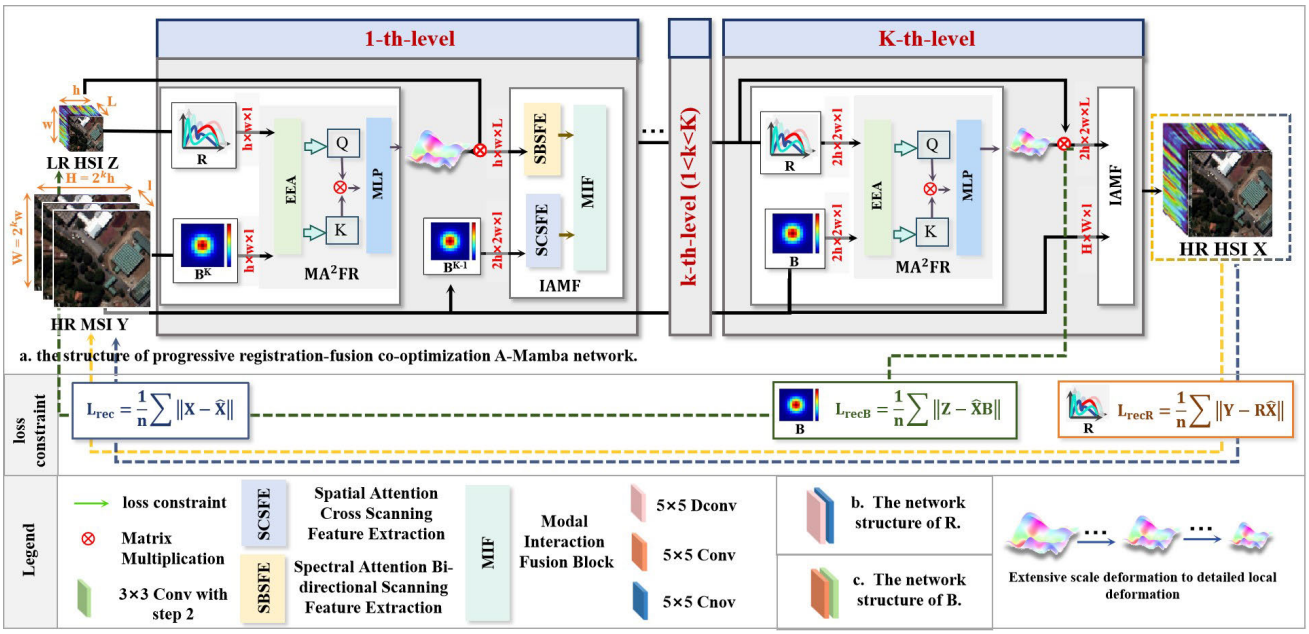


Fig. 3. Overall architecture of the proposed PRFCoAM. (a) Structure of PRFCoAM. (b) Network structure of $\boldsymbol{R}$. (c) Network structure of $\boldsymbol{B}$.

two independent steps, which leads to the accumulation of errors generated during the registration and subsequent fusion.

Therefore, in this article, we integrated registration and fusion into an MAPRF to achieve progressive registration-fusion co-optimization, and the MAPRF integrates MULAR and IAMF. By iterating MAPRF K levels, the proposed network gradually corrects deformations from extensive to detailed and refines the fusion result level by level. MULAR learns the spectral response function and the PSF so that the inputs can be aligned under a uniform reference standard then enhances the edge information of the image to help align the spatial pixels of local regions according to correlation. IAMF scans the well-registered image in multiple directions and implants an attention block for adaptive learning of the features most conducive to fusion. This process can be expressed as

$$\mathbf{X}^{(k+1)} = \mathcal{F}^{(k)}\left(\mathcal{D}^{(k)}, \ldots, \mathcal{F}^{(1)}\left(\mathcal{D}^{(1)}\left(\mathbf{Y}, \mathbf{X}^{(1)}\right) \otimes \mathbf{X}^{(1)}, \mathbf{Y}\right)\right) \quad 1 \le k \le K; \mathbf{X}^{(1)} = \mathbf{Z} \tag{4}$$

where $\mathcal{D}^{(k)}(\cdot)$ represents the kth level of MULAR and $\mathcal{F}^{(k)}$ represents the kth level of IAMF.

B. Overview

As shown in Fig. 3, the proposed PRFCoAM consists of AMRF and achieves registration-fusion co-optimization from an extensive to a detailed level. By performing MAPRF K levels, PRFCoAM progressively corrects the nonrigid pixel offset and refines the fusion results at each level, obtaining the desired HR HSI. MAPRF integrates MULAR and IAMF at each level to achieve registration-fusion co-optimization. MULAR adaptively aligns the modality of the input images through $\boldsymbol{R}$ and $\boldsymbol{B}$ from an extensive to a detailed level, then enhances the feature of edge to assist the further registration. With this assistance, MULAR corrects nonrigid pixel offset in smaller neighborhood ranges by embedding the local attention that could focus on the correlations and differences between corresponding regions of input images. The well-registered images are input to the IAMF for further fusion. IAMF integrates spatial attention-enhanced cross-scanning feature extraction (SCSFE), spectral attention-enhanced bidirectional scanning feature extraction (SBSFE), and multimodal interactive fusion (MIF). SCSFE and SBSFE extract spatial and spectral features by multidirectional scanning and bidirectional scanning, respectively, overcoming the dilemma of the unidirectional information induction caused by Mamba in extracting continuous spatial and spectral features. They embed the learnable attention block in the SSM, guiding the IAMF to grasp the most critical spatial and spectral features that are most conducive to fusion. MIF integrates the complementary information from SBSFE and SCSFE and avoids the omission of subtle but critical features, to obtain the desired HR HSI. The output $\mathbf{X}^{(k+1)}$ of the kth level MAPRF in PRFCoAM is obtained as follows:

$$\mathbf{X}^{(k+1)} = \mathcal{F}_{\mathrm{A-M}}^{(k)}\left(\mathcal{D}^{(k)}\left(\mathbf{YB}^{(K-k+1)}, \mathbf{RX}^{(k)}\right) \otimes \mathbf{X}^{(k)}, \mathbf{YB}^{(K-k)}\right) \quad 1 \le k \le K; \mathbf{X}^{(1)} = \mathbf{Z} \tag{5}$$

where $\mathcal{D}^{(k)}(\cdot)$ represents the kth level of MULAR and $\mathcal{F}_{\mathrm{A-M}}^{(k)}(\cdot)$ represents the kth level of IAMF.

C. MULAR Block

Due to differences in imaging conditions, LR HSIs and HR MSIs typically cannot achieve effective spatial alignment. Affine alignment methods for large-scale remote sensing images can easily obtain full-frame registration of HSIs and MSIs. However, it is challenging to handle local nonlinear deformation, such as the deformation of a local region or a single object. Therefore, as shown in Fig. 4, the MULAR block is proposed that embeds the local attention to focus on the correlations and differences between corresponding regions, thus

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:13:02 UTC from IEEE Xplore. Restrictions apply.

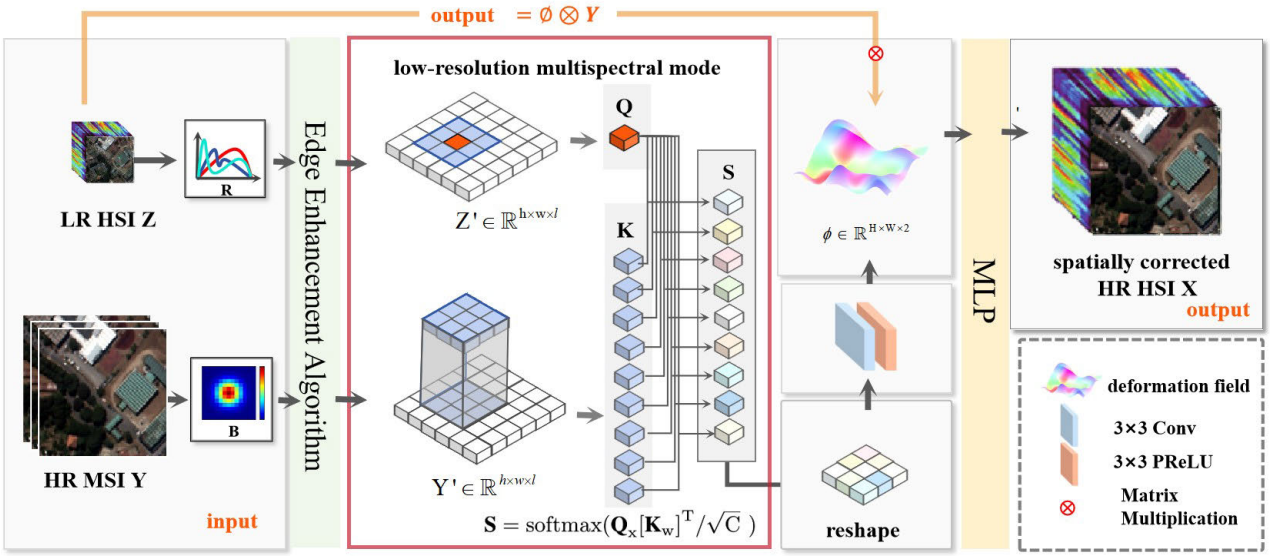


Fig. 4. Overall architecture of the proposed MULAR.

correcting nonlinear deformation from extensive to detailed. The spatially corrected LR HSI $\mathbf{X}^{(k)}$ of the MULAR output can be expressed as follows:

$$\mathbf{X}^{(k)} = \mathcal{D}^{(k)}(\mathbf{YB}^{(K-k+1)}, \mathbf{RX}^{(k)}), \quad 1 \le k \le K;\ \mathbf{X}^{(1)} = \mathbf{Z}. \tag{6}$$

The learning process of $\boldsymbol{R}$ and $\boldsymbol{B}$ is restricted in the loss function to ensure that the learned model has a clear physical meaning, that is, effective spectral and spatial degradation is achieved without pretraining.

Specifically, due to the difference in imaging equipment and principles, HSIs and MSIs exhibit spatial and spectral differences that increase the difficulty of registration. Modal transformation serves to diminish differences between them and to help determine the corresponding relationships between feature points or regions more accurately. Therefore, it is necessary to transform the input of each level into the unified modality to assist in registration and perform spatial and spectral downsampling on LR HSI $\mathbf{Z}$ and HR MSI $\mathbf{Y}$ at the kth level, respectively. Obtained the unified modality $\mathbf{Z}\downarrow_{\text{spe}}^{(k)}$ and $\mathbf{Y}\downarrow_{\text{spa}}^{(k)}$ can be expressed as follows:

$$\begin{aligned} \mathbf{Z}{\downarrow_{\text{spe}}}^{(k)} &= \text{Conv}_{\boldsymbol{R}}(\mathbf{X}^{(k)}) \\ \mathbf{Y}{\downarrow_{\text{spa}}}^{(k)} &= \text{Conv}_{\boldsymbol{B}}(\mathbf{Y}^{(k)}), \quad 1 \le k \le K; \mathbf{X}^{(1)} = \mathbf{Z} \end{aligned} \tag{7}$$

where $\text{Conv}_{\boldsymbol{R}}(\cdot)$ denotes the convolution layers that learn the spectral degradation function and $\text{Conv}_{\boldsymbol{B}}(\cdot)$ denotes the convolution layers that learn the spatial transformation matrix.

Edges located in areas of intense change are less sensitive to variations in light and color than texture features. Therefore, the features of the edge are more stable under different conditions. In addition, edge details are usually present in the high-frequency components of the spectrum, so the input image $\mathbf{Z}\downarrow_{\text{spe}}^{(k)}$ at the kth level is transformed into the frequency domain using the discrete cosine transform (DCT), where low-frequency components are suppressed to enhance edge details and facilitate registration. As shown in Fig. 5, the

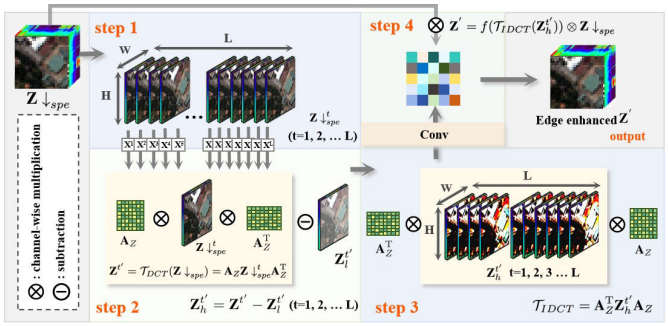


Fig. 5. Process of the edge enhancement algorithm.

features $(\mathbf{Z}^{t'})^{(k)}$ at the kth level in the frequency domain obtained by the DCT algorithm are expressed as follows:

$$(\mathbf{Z}^{t'})^{(k)} = \mathcal{T}_{\text{DCT}}(\mathbf{Z}\downarrow_{\text{spe}}{}^{(k)}) = \mathbf{A}_Z\left(\mathbf{Z}\downarrow_{\text{spe}}^{t}\right)^{(k)}\mathbf{A}_Z^{\text{T}}, \quad 1 \le k \le K; \quad t = 1, 2, \ldots, L \tag{8}$$

where $\mathcal{T}_{\text{DCT}}(\cdot)$ represents the DCT operation, $\mathbf{A}_Z$ represents the transformation matrix, $(\cdot)^{\text{T}}$ represents the transpose operation, and L is the number of spectral bands of $\mathbf{Z}\downarrow_{\text{spe}}^{(k)}$.

The inverse DCT (IDCT) is used to reverse the frequency domain into the spatial domain without losing information or damaging other parts of the image quality. The low-frequency features $(\mathbf{Z}_l^{t'})^{(k)}$ of $(\mathbf{Z}^{t'})^{(k)}$ at the kth level are suppressed in the frequency domain, and the guidance of the high-frequency detail $(\mathbf{Z}_h^{t'})^{(k)}$ at the kth level is implicitly learned and transformed in the spatial domain by IDCT. The obtained $(\mathbf{Z}')^{(k)}$ at the kth level is represented as

$$\begin{aligned} (\mathbf{Z}')^{(k)} &= f\left(\mathcal{T}_{\text{IDCT}}\left(\left(\mathbf{Z}_h^{t'}\right)^{(k)}\right)\right) \otimes \mathbf{Z}\downarrow_{\text{spe}}{}^{(k)} \\ &= f(\mathcal{T}_{\text{IDCT}}\left((\mathbf{Z}^{t'})^{(k)} - \left(\mathbf{Z}_l^{t'}\right)^{(k)}\right) \otimes \mathbf{Z}\downarrow_{\text{spe}}{}^{(k)} \\ &\qquad 1 \le k \le K; \quad t = 1, 2, \ldots, L \end{aligned} \tag{9}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:13:02 UTC from IEEE Xplore. Restrictions apply.

where $\mathcal{T}_{\mathrm{IDCT}}(\cdot)$ represents the IDCT algorithm, $\mathbf{Z}_h^{t'}$ is the high-frequency feature of the tth band in $\mathbf{Z}^{t'}$, $\mathbf{Z}_l^{t'}$ is the low-frequency feature of the tth band in $\mathbf{Z}^{t'}$, and the convolution kernel size of $f(\cdot)$ is set to equal the scale of the image. Similar to $\mathbf{Z}\downarrow_{\mathrm{spe}}^{(k)}$, we enhance the edge details of $\mathbf{Y}\downarrow_{\mathrm{spa}}^{(k)}$ to obtain $(\mathbf{Y}')^{(k)}$.

After obtaining the detail-enhanced image, the local attention mechanism is implanted to densely register the spatial pixels and repair the nonrigid pixel offset in a unified modality. Specifically, we define $\mathbf{Z}[i, j]$ as the pixel value in image $\mathbf{Z}'$ with coordinate (i, j), $\mathbf{Y}[m, n]$ as the pixel value in image $\mathbf{Y}'$ with the coordinate (m, n), and $\mathbf{W}[m, n]$ as the set of values in the image with the coordinate (m, n) as the center pixel including the surrounding w adjacent pixels. Since the input image pair has been coarsely registered, we select a smaller range of matching pixels (neighborhood size $w = 3$). $\mathbf{Z}[i, j]$ and $\mathbf{W}[m, n]$ are mapped through a linear layer to produce two different vectors, which are the query vector $\mathbf{Q}$ (query) and the key vector $\mathbf{K}$ (key)

$$\mathbf{Q}_x = \mathrm{linear}_x(\mathbf{Z}[i, j]) \tag{10}$$

$$\mathbf{K}_w = \mathrm{linear}_w(\mathbf{W}[m, n]) \tag{11}$$

where $\mathrm{linear}_x(\cdot)$ represents the mapping of $\mathbf{Z}[i, j]$ to obtain $\mathbf{Q}_x$ and $\mathrm{linear}_w(\cdot)$ represents the mapping of $\mathbf{W}[m, n]$ to obtain $\mathbf{K}_w$. To obtain the correlation between the current point and the surrounding pixels in the input feature map, the cross-attention mechanism is used to capture the correspondence between HSIs and MSIs and to calculate the spatial correlation pixel by pixel. The obtained offset matrix $\mathbf{S}$ is represented as

$$\mathbf{S} = \mathrm{softmax}(\mathbf{Q}_x[\mathbf{K}_w]^{\mathrm{T}}/\sqrt{\mathbf{C}}\ ) \tag{12}$$

where $\mathbf{C}$ represents the matrix dimension of $\mathbf{K}_w$.

The offset matrix $\mathbf{S}$ is input into a learnable linear convolution layer to generate a spatial deformation field, which is a spatial transition matrix with the size $H \times W \times 2$. The two spatial dimensions correspond to the displacement of the x-axis and the y-axis of each pixel, respectively, such that x and y correspond to the most matched positions. The deformation field ϕ is represented as follows:

$$\phi = f_D(\mathbf{S}) \tag{13}$$

where $f_D(\cdot)$ is the convolution layer that learns space transformation parameters. The corrected HSI image $\mathbf{Z}_{\mathrm{cor}}^{(k)}$ at the kth level can be obtained as

$$\mathbf{Z}_{\mathrm{cor}}^{(k)} = \phi \otimes \mathbf{Z}^{(k)} \tag{14}$$

where $\phi \otimes \mathbf{Z}$ represents the deformation field ϕ acts on $\mathbf{Z}$.

D. IAMF Block

1) Preliminaries: The SSM derived from linear time-invariant (LTI) systems in modern control theory which maps a 1-D input sequence signal $x(t) \in \mathbb{R}^N$ to output and $y(t) \in \mathbb{R}^N$ through a latent space $h(t) \in \mathbb{R}^N$. This model uses the state transition matrix $\mathbf{A} \in \mathbb{R}^{N\times N}$ and the projection matrices $\boldsymbol{B} \in \mathbb{R}^{N\times 1}$ and $\mathbf{C} \in \mathbb{R}^{N\times 1}$ to define linear ordinary differential equations (ODEs), which is described as

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \boldsymbol{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{15}$$

where N is the state size.

To accommodate DL, the continuous system described above usually employs a discretization process. Specifically, a timescale parameter Δ transforms the continuous parameters $\mathbf{A}$ and $\boldsymbol{B}$ to the discrete counterparts $\overline{\mathbf{A}}$ and $\overline{\boldsymbol{B}}$. A prevalent way for discretization is to use the zero-order holder (ZOH)

$$\begin{aligned} \overline{\mathbf{A}} &= \exp(\Delta\mathbf{A}) \\ \overline{\boldsymbol{B}} &= \Delta\mathbf{A}^{-1}\exp(\Delta\mathbf{A} - I)\cdot\Delta\boldsymbol{B}. \end{aligned} \tag{16}$$

After the discretization of $\overline{\mathbf{A}}$ and $\overline{\boldsymbol{B}}$, the ordinary differential (15) of the discretized SSM is written as

$$\begin{aligned} h_t &= \overline{\mathbf{A}}h_{t-1} + \boldsymbol{B}x_t \\ y_t &= \mathbf{C}h_t. \end{aligned} \tag{17}$$

The discretization described above provides efficiency. However, parameters in SSM are data-independent and LTI, which limits the efficiency and expressiveness of hidden states in context compression. To mitigate this problem, Mamba selectively scans the sequential state for interactions and makes the parameters dependent on the input. This recursive process can be formulated as a convolution layer

$$\begin{aligned} \boldsymbol{B}_t, \mathbf{C}_t, \Delta_t = {} & \mathrm{Linear}_{\boldsymbol{B}}[x(t)], \mathrm{Linear}_c[x(t)] \\ & \mathrm{SoftPlus}[\Delta + \mathrm{Linear}_{\Delta}[x(t)]]. \end{aligned} \tag{18}$$

2) Spatial Attention-Enhanced Cross-Scanning Feature Extraction: Some pixels that are spatially close in 2-D feature maps may become distant from each other after being transformed into a 1-D token sequence, which potentially leads to the loss of local information. Therefore, as shown in Fig. 6, the proposed SCSFE introduces a selective SSM mechanism in spatial dimension to capture the spatial long-term dependencies of MSIs. Considering that the SiLU/Swish activation function in Mamba simply relies on the feature intensity to obtain the gating coefficient, the proposed SCSFE adopts a spatial attention map as the gating coefficient which learns the spatial relationships between features. The spatial attention block can help focus on the critical spatial regions for further improving fusion while suppressing the less important areas.

The proposed SCSFE scans the spatial feature in four different directions, from top to bottom and left to right, and vice versa, so that each pixel integrates information from all the adjacent pixels in four directions. Specifically, we first flatten the given patch $\mathbf{S} \in \mathbb{R}^{M^2\times l}$ of MSI ($\mathbf{Y}$) into a 1-D sequence $\mathcal{S}_i = \{[\mathbf{s}_0^i, \mathbf{s}_1^i, \ldots, \mathbf{s}_{M\times M}^i]|\mathbf{s}_j^i \in \mathbb{R}^{1\times l};\ \mathrm{i} \in (0, 1, 2, 3)\}$ by preset route and recursively calculate each sequence

$$\begin{aligned} \mathbf{h}_j^i &= \overline{\mathbf{A}}_{\mathrm{ms}}\mathbf{h}_{j-1}^i + \overline{\boldsymbol{B}}_{\mathrm{ms}}\mathbf{s}_{j1}^i \\ \mathbf{y}_j^i &= \mathbf{C}_{\mathrm{ms}}\overline{\mathbf{h}}_j^i + \mathbf{s}_{j1}^i \end{aligned} \tag{19}$$

where $\overline{\mathbf{A}}_{\mathrm{ms}}$, $\overline{\boldsymbol{B}}_{\mathrm{ms}}$, and $\overline{\mathbf{C}}_{\mathrm{spa}}$ represent the trainable parameters in SCSFE. The output sequence $\mathcal{Y}_{\mathrm{spa}} = \{[\mathbf{y}_0^i, \mathbf{y}_1^i, \ldots, \mathbf{y}_{M\times M}^i]|\mathbf{y}_j^i \in \mathbb{R}^{1\times l};\ i \in (0, 1, 2, 3)\}$ of the four scan directions is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:13:02 UTC from IEEE Xplore. Restrictions apply.

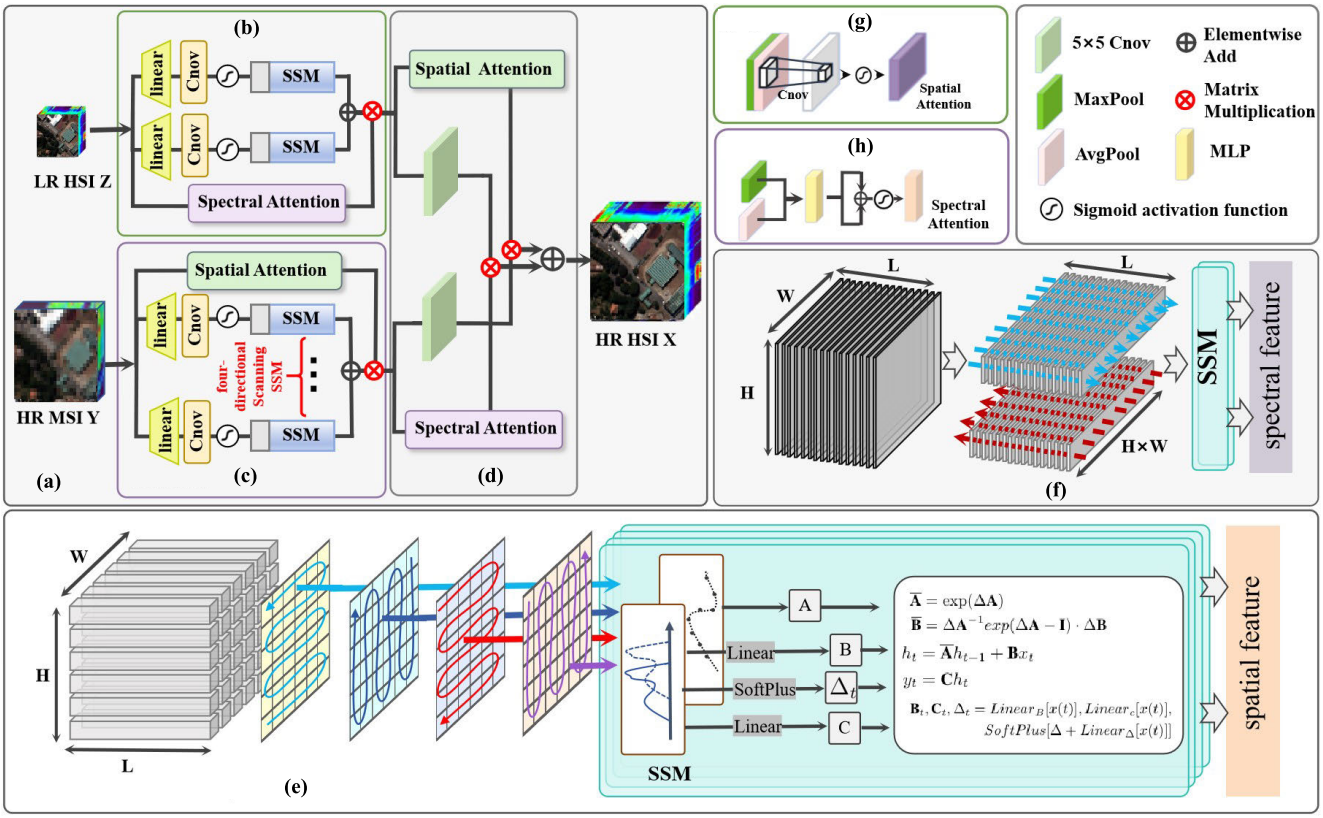


Fig. 6. Overall architecture of the proposed IAMF block. (a) IAMF. (b) SBSFE. (c) SCSFE. (d) MIF. (e) Four-directional scanning SSM in SCSFE. (f) Bi-directional scanning SSM in SBSFE. (g) Spatial attention. (h) Spectral attention.

then fused so that each element in the output sequence can integrate information of its neighboring regions. To grasp the most important spatial information, the spatial attention block redistributes the weight of the features extracted by SSM scanning in four directions to extract the most critical fusion information and reduce the attention to useless features. The output $\mathbf{F}_{\text{msi}} \in \mathbb{R}^{H\times W\times l}$ of SCSFE at the kth level of PRFCoAM can be expressed as

$$\mathbf{F}_{\text{msi}} = \mathcal{Y}_{\text{spa}} \otimes f_{\text{att-msi}}(\mathbf{Y} \cdot \boldsymbol{B}^{(K-k)}) \tag{20}$$

where $f_{\text{att-msi}}(\cdot)$ is the convolution layers in spatial attention.

3) Spectral Attention-Enhanced Bi-Directional Scanning Feature Extraction: Mamba induces unidirectional information flow, sequentially transmitting and processing features. Specifically, when Mamba processes input information, the sequence flows unidirectionally from one state to the next without a feedback mechanism. However, HSIs have a strong spectral correlation, so the one-way sensing mechanism of the original Mamba is not sufficient to analyze HSIs. To solve this problem, the spectral attention-enhanced bidirectional scanning feature extraction (SBSFE) proposed in this article scans the spectral channels in both directions, effectively integrating the spectral features and capturing the dependency across channels of HSIs. To effectively concentrate on the more important features, we replace the SiLU activation function with spectral attention. The learned spectral attention gives new weights to the extracted spectral features, thus filtering out irrelevant information and improving the fusion.

Specifically, patch $\mathbf{P} \in \mathbb{R}^{N^2\times L}$ of the HSI ($\mathbf{Z}_{\text{cor}}$) is first flattened into 1-D sequence $\mathcal{P}_m = \{[\mathbf{p}_0^m, \mathbf{p}_1^m, \ldots, \mathbf{p}_L^m] | \mathbf{p}_n^m \in \mathbb{R}^{1\times N^2}; m \in (0, 1)\}$ along the spatial dimension, and the output sequence $\mathcal{Y}_{\text{spe}} = \{[\overline{\mathbf{y}}_0^m, \overline{\mathbf{y}}_1^m, \ldots, \overline{\mathbf{y}}_L^m] | \overline{\mathbf{y}}_n^m \in \mathbb{R}^{1\times N^2}; m \in (0, 1)\}$ is obtained by scanning $\mathcal{P}_m$ from two directions. The calculation process is shown as follows:

$$\begin{aligned} \overline{\mathbf{h}}_n^m &= \overline{\mathbf{A}}_{\text{hsi}} \overline{\mathbf{h}}_{n-1}^m + \overline{\boldsymbol{B}}_{\text{hsi}} \mathbf{s}_{n1}^m \\ \overline{\mathbf{y}}_n^m &= \mathbf{C}_{\text{hsi}} \mathbf{h}_n^m + \mathbf{s}_{n1}^m \end{aligned} \tag{21}$$

where $\overline{\mathbf{A}}_{\text{hsi}}$, $\overline{\boldsymbol{B}}_{\text{hsi}}$, and $\overline{\mathbf{C}}_{\text{hsi}}$ represent the trainable parameters. Then, the bidirectional output sequence is fused so that each element of the output sequence can integrate the information of adjacent bands. By reweighting the feature $\mathcal{Y}_{\text{spe}}$ with spectral attention, the output $\mathbf{F}_{\text{hsi}} \in \mathbb{R}^{h\times w\times L}$ of SBSFE at the kth level of PRFCoAM can be expressed as

$$\mathbf{F}_{\text{hsi}} = \mathcal{Y}_{\text{spe}} \otimes f_{\text{att-hsi}}(\boldsymbol{R} \cdot \mathbf{Z}_{\text{cor}}) \tag{22}$$

where $f_{\text{att-hsi}}(\cdot)$ stands for convolution layers in spectral attention.

4) Multimodal Interaction Fusion: To avoid the omission of SBSFE and SCSFE in extracting the subtle but critical features, MIF interactive embedding spatial and spectral attention blocks to integrate the complementary features of SBSFE and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:13:02 UTC from IEEE Xplore. Restrictions apply.

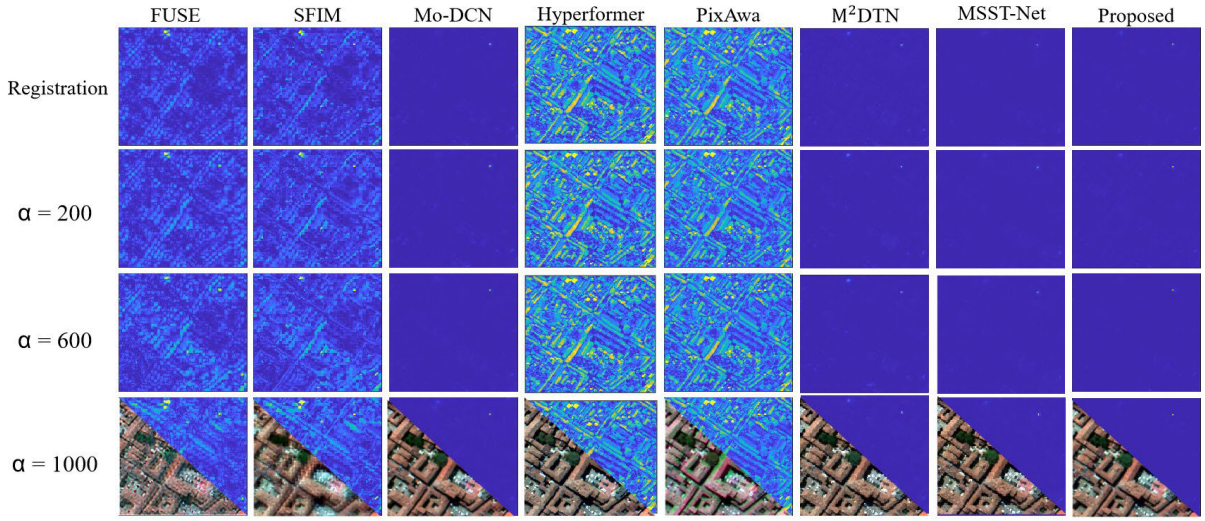


Fig. 7. Visual comparison of results of different methods on the Pavia Centre dataset with varying degrees of elastic distortion.

SCSFE, the obtained features $\mathbf{Y}_{\text{fus}}$ of the desired HR HSI

$$\mathbf{Y}_{\text{fus}} = f_{\text{exa}}(F_{\text{msi}}) * f_{\text{att-ms}}(F_{\text{hsi}}) + f_{\text{exa}}(F_{\text{hsi}}) * f_{\text{att-hsi}}(F_{\text{msi}}) \tag{23}$$

where $f_{\text{exa}}(\cdot)$ represents the convolution layers in MIF.

E. Network Training

1) HR HSI Reconstruction: In this experiment, the similarity of the reconstructed HR HSI and the reference HR HSI is measured using the $L1$ norm as a metric function

$$L_{\text{rec}} = \frac{1}{n}\sum_{i=1}^{n} \|\mathbf{X} - \hat{\mathbf{X}}\|_1 \tag{24}$$

where n is the number of images in a training batch, $\mathbf{X}$ is the ground truth, and $\hat{\mathbf{X}}$ is the output of the network. To ensure that the learned $\boldsymbol{B}$ has a clear physical meaningful, the LR HSI and output are constrained by the $L1$ loss function, which is defined as follows:

$$L_{\text{rec}\boldsymbol{B}} = \frac{1}{n}\sum_{i=1}^{n} \|\mathbf{Z} - \hat{\mathbf{X}}\boldsymbol{B}\|_1. \tag{25}$$

Similar to $\boldsymbol{B}$, the $L1$ loss function constrains the physical meaning of $\boldsymbol{R}$, which can be defined as follows:

$$L_{\text{rec}\boldsymbol{R}} = \frac{1}{n}\sum_{i=1}^{n} \|\mathbf{Y} - \boldsymbol{R}\hat{\mathbf{X}}\|_1. \tag{26}$$

2) Deformation Smoothing: In this experiment, deformation field smoothing loss L_{smooth} is introduced to avoid the problem of insufficient smoothness of the deformation field

$$L_{\text{smooth}} = \sum_{i,j} \|\nabla u_{ij}\|^2 \tag{27}$$

where u_{ij} represents the deformation vector of the deformation field at pixel position (i, j) and ∇ denotes the gradient operator.

Finally, the overall loss function of the network is defined as

$$L = \alpha L_{\text{rec}} + \beta_1 L_{\text{rec}\boldsymbol{B}} + \beta_2 L_{\text{rec}\boldsymbol{R}} + \gamma L_{\text{smooth}} \tag{28}$$

where $\alpha 1$, $\beta 1$, $\beta 2$, and γ are the tradeoff parameters.

III. Experimental Results and Analysis

A. Data Description and Experimental Setup

1) Simulated Datasets: The Pavia Centre (PaviaC), Houston, and Chikusei datasets are selected to verify the effectiveness and generalizability of the proposed PRFCoAM.

1) The PaviaC dataset, captured by the ROSIS sensor over Pavia, Italy, in the wavelength range of 430–860 nm, consists of a 960 × 640 × 102 image. After excluding 13 absorbent and noise bands, 102 bands remain for use. This image is divided into 12 nonoverlapping patches, and nine patches are randomly chosen as the training set, while the remaining three are used for testing.
2) The Houston dataset, captured by the ITRES-CASI 1500 HS sensor, covers the University of Houston campus and nearby urban areas within the 380–1050-nm wavelength range. The dataset is divided into eight nonoverlapping patches, with six patches used for training, and the remaining two used for testing.
3) The Chikusei dataset was obtained by the Headwall Hyperspec-VNIR-C sensor on July 29, 2014, in Chikusei, Japan. The HSI contains 2517 × 2335 pixels and 128 bands with a band range of 343–1018 nm and a spatial resolution of 2.5 m.

2) Simulated Details: For all simulated LR HSIs, we employ elastic transformation [33], homography transformation [34], and turbulence transformation [35] to simulate the distortions that may occur in the image acquisition process in reality. To better evaluate the adaptability of PRFCoAM to both slight and extreme deformations, as well as to investigate its generalization ability, we set different deformation degrees for LR HSIs from small to large. Specifically, α, which is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:13:02 UTC from IEEE Xplore. Restrictions apply.

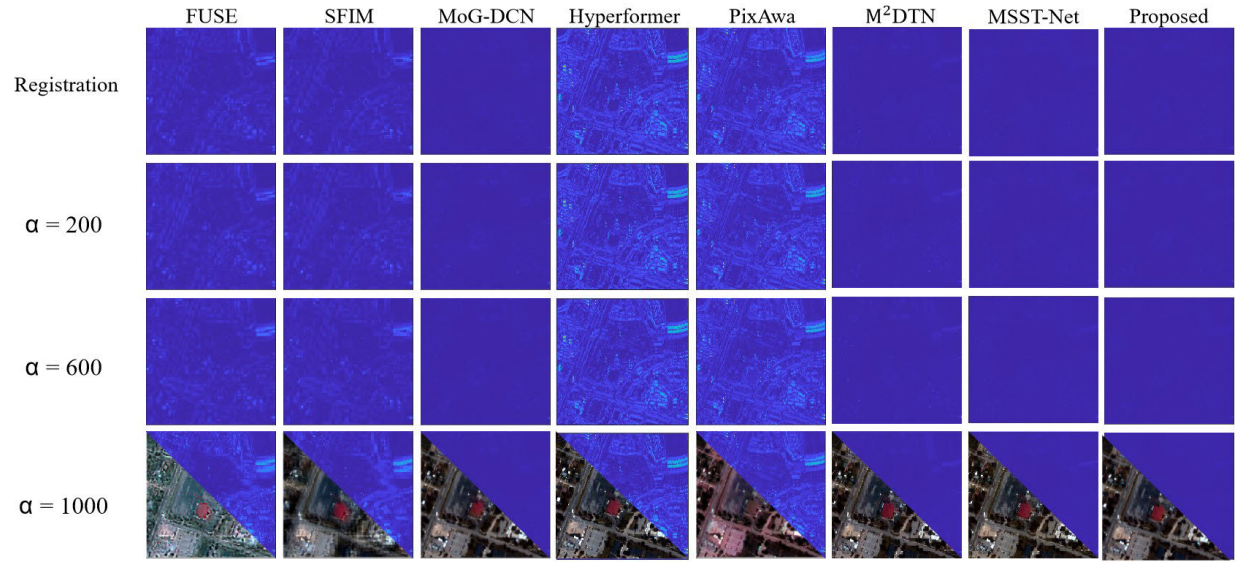


Fig. 8. Visual comparison of results of different methods on the Houston dataset with varying degrees of elastic distortion.

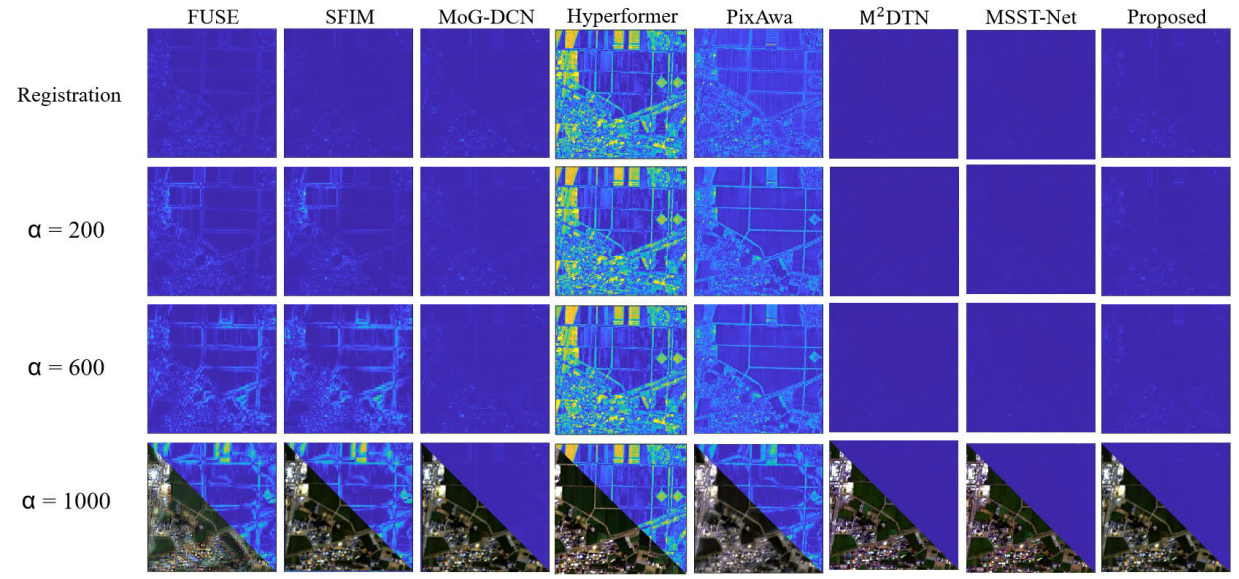


Fig. 9. Visual comparison of results of different methods on the Chikusei dataset with varying degrees of elastic distortion.

set to 200, 600, and 1000, acts as a factor that controls the strength of deformation. For the Pavia Centre datasets, we simulate homography transformation and turbulence transformation. The maximum pixel offset (MPO) for homography transformation is set to 2, and the ratio r of aperture diameter D and fried parameter r_0 for turbulence transformation is set to 8.

3) Real Datasets: We assess the effectiveness of the proposed method on a real dataset (XijingU dataset). The XijingU dataset was collected by our team at Xijing University, located in Xi'an, Shaanxi, China. It comprises HSIs captured using the Headwall VNIR/SWIR Full Band HSIs and MSIs, obtained via the FLIR Duo Pro ***R*** sensor. After processing, the sizes of the HSI and MSI were 960 × 960 × 137 and 3840 × 3840 × 3, respectively. As ground-truth data was unavailable, we adopted Wald's protocol to generate the dataset for supervised learning. Specifically, Wald's protocol is employed to process the observed HSI, resulting in LR HSIs of size 240 × 240 × 137 and the MSI of size 960 × 960 × 3, respectively. To enlarge the training dataset, 69 image pairs with sizes of 40 × 40 and 160 × 160 were generated by cropping the overlapping training samples.

4) Evaluation Measures: For the simulated datasets, six widely used metrics are employed to quantitatively evaluate the performance of the images, including the peak SNR (PSNR), correlation coefficients (CCs), spectral angle mapper (SAM), root mean square error (RMSE), and error relative globale adimensionnelle de synthèse (ERGAS). Since the real dataset had no reference images available, the quality with no reference (QNR) index [36] is employed for quality assessment, QNR closer to 1 indicates better fusion quality.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:13:02 UTC from IEEE Xplore. Restrictions apply.

TABLE I

QUANTITATIVE COMPARISON ON THREE SIMULATED DATASETS WITH VARYING DEGREES OF ELASTIC DEFORMATION (COLOR CONVENTION: **BEST** AND SECOND-BEST)

Degrees of distortion	Index	Traditional methods		DL-based methods					
		FUSE	SFIM	MoG-DCN	Hyperformer	pixAwa	M^2DTN	MSST-Net	Ours
Pavia Center									
well-registration	PSNR	28.8160	28.4004	42.2127	35.2354	30.4081	41.535	42.8875	**43.6771**
	CC	0.9155	0.9039	0.9888	0.9811	0.9390	0.9886	0.9891	**0.9920**
	SAM	7.9479	5.2229	3.0291	7.3114	8.2797	3.1978	2.9451	**2.6607**
	RMSE	0.0363	0.0381	0.0079	0.0168	0.0302	0.0085	0.0073	**0.0066**
	ERGAS	5.5733	5.9623	1.8147	2.9603	4.8526	1.8324	1.7307	**1.5100**
$\alpha = 200$	PSNR	28.8182	28.6433	42.0465	36.0340	29.4331	41.8963	42.4549	**43.4868**
	CC	0.9113	0.9057	0.9887	0.9821	0.9313	0.9885	0.9886	**0.9909**
	SAM	7.3113	5.3734	3.0786	6.7297	9.0148	3.1551	3.1376	**2.7550**
	RMSE	0.0365	0.0374	0.0081	0.0159	0.0338	0.0082	0.0076	**0.0068**
	ERGAS	5.6446	5.8776	1.8524	2.8050	5.2597	1.8355	1.7761	**1.5781**
$\alpha = 600$	PSNR	26.2939	25.8335	42.1068	35.7501	31.9575	41.5617	42.1964	**43.3218**
	CC	0.8368	0.8063	0.9890	0.9821	0.9569	0.9879	0.9882	**0.9909**
	SAM	9.6801	6.8529	3.1736	6.7883	7.2680	3.2260	3.2126	**2.8081**
	RMSE	0.0494	0.0520	0.0080	2.8262	0.0253	0.0085	0.0079	**0.0069**
	ERGAS	7.7374	8.1443	1.7982	**0.0159**	4.2080	1.8945	1.8044	1.5789
$\alpha = 1000$	PSNR	23.6112	23.3650	41.6752	35.6738	26.4920	41.1655	41.8897	**43.2514**
	CC	0.7418	0.6709	0.9886	0.9817	0.8511	0.9879	0.9883	**0.9912**
	SAM	13.9250	8.3822	3.2323	6.0510	13.0642	3.3213	3.2724	**2.8062**
	RMSE	0.0705	0.0809	0.0084	2.8477	0.0478	0.0088	0.0082	**0.0070**
	ERGAS	11.1187	11.1842	1.8276	**0.0161**	7.6707	1.9171	1.8363	1.5611
Houston									
well-registration	PSNR	33.4135	36.7272	50.8053	38.9857	35.5415	52.1444	52.6801	**52.7245**
	CC	0.8721	0.9442	0.9442	0.9835	0.8998	0.9934	0.9946	**0.9953**
	SAM	7.1585	3.8122	1.2602	2.3292	6.1546	1.1075	**1.1051**	1.1305
	RMSE	0.0214	0.0146	0.0029	0.0059	0.0167	0.0025	**0.0023**	**0.0023**
	ERGAS	5.3674	3.7886	1.0171	1.9400	4.4704	1.0011	0.8805	**0.8571**
$\alpha = 200$	PSNR	33.2447	36.7490	50.2992	38.9423	36.5867	49.4327	52.4856	**52.6818**
	CC	0.8686	0.9442	0.9933	0.9849	0.9198	0.9911	0.9943	**0.9950**
	SAM	7.4889	3.9985	1.3849	2.4919	5.5997	1.5458	1.1158	**1.0718**
	RMSE	0.0218	0.0146	0.0031	0.0062	0.0148	0.0034	0.0024	**0.0023**
	ERGAS	5.4802	3.7837	1.0519	1.8159	3.8564	1.3332	0.9409	**0.8697**
$\alpha = 600$	PSNR	32.0558	33.9982	49.6969	37.9050	35.8830	49.9423	51.4002	**52.4692**
	CC	0.8186	0.8888	0.9926	0.9850	0.9182	0.9908	0.9933	**0.9948**
	SAM	8.4899	5.5374	1.4577	2.5642	5.7019	1.3730	1.2140	**1.0718**
	RMSE	0.0251	0.0203	0.0033	0.0070	0.0161	0.0032	0.0027	**0.0024**
	ERGAS	6.3159	5.2180	1.1309	2.0367	4.0823	1.2523	1.0305	**0.8870**
$\alpha = 1000$	PSNR	31.4010	32.0448	49.4391	37.0342	34.2072	49.693	50.8782	**52.3138**
	CC	0.7649	0.8217	0.9923	0.9802	0.8852	0.9906	0.9925	**0.9945**
	SAM	9.3380	6.9488	1.5026	3.1653	6.5910	1.4311	1.2535	**1.1325**
	RMSE	0.0282	0.0254	0.0034	0.0076	0.0195	0.0033	0.0029	**0.0024**
	ERGAS	7.1154	6.5103	1.1409	2.4115	5.0360	1.3723	1.0944	**0.9184**
Chikusel									
well-registration	PSNR	39.2138	51.4854	50.1714	38.0156	34.1434	51.8206	52.2532	**52.6759**
	CC	0.9385	0.9923	0.9873	0.9853	0.8439	0.9887	0.9911	**0.9924**
	SAM	3.1494	1.3730	1.5639	1.7710	7.1167	1.3318	1.2611	**1.1589**
	RMSE	0.0112	0.0031	0.0032	0.0046	0.0199	0.0026	0.0025	**0.0024**
	ERGAS	6.0442	**1.8690**	2.5221	2.9123	9.0570	2.3278	2.1691	2.1451
$\alpha = 200$	PSNR	38.0568	45.1507	50.1293	36.8904	33.7634	52.0779	51.9590	**52.6867**
	CC	0.9229	0.9832	0.9895	0.9833	0.8439	0.9887	0.9904	**0.9926**
	SAM	3.5652	1.7853	1.6305	2.0127	6.8949	1.2820	1.3144	**1.1457**
	RMSE	0.0128	0.0057	0.0032	0.0047	0.0208	0.0026	0.0025	**0.0024**
	ERGAS	6.7290	2.9260	2.4158	3.2001	13.0818	2.5131	2.1572	**2.1169**
$\alpha = 600$	PSNR	35.7209	38.2493	49.6074	36.6103	33.6842	51.5563	51.3971	**52.4994**
	CC	0.5673	0.9318	0.9882	0.9821	0.8298	0.9886	0.9893	**0.9920**
	SAM	4.3780	2.9893	1.7116	2.2341	7.0839	1.2936	1.3774	**1.1678**
	RMSE	0.0167	0.0126	0.0033	0.0050	0.0210	0.0027	0.0027	**0.0024**
	ERGAS	8.3979	5.8998	2.5173	3.2321	13.6809	2.3650	2.2464	**2.2266**
$\alpha = 1000$	PSNR	30.5460	30.6296	49.6177	36.7728	31.8696	51.3559	49.624	**52.2895**
	CC	0.5993	0.6456	0.9872	0.9832	0.7926	0.9884	0.9869	**0.9909**
	SAM	7.3429	6.7342	1.7086	1.9080	8.7555	1.3240	1.6377	**1.2898**
	RMSE	0.0304	0.0302	0.0034	0.0054	0.0260	0.0027	0.0034	**0.0025**
	ERGAS	13.4786	13.0040	2.6494	3.0523	10.4247	2.4063	2.3590	**2.2449**

B. Comparisons With State-of-the-Art Methods

To fully verify the effectiveness of PMI-RFCoNet, we compared it with fast fusion based on Sylvester equation (FUSE) [37], SFIM [38], MoG-DCN [39], hyperformer [40], PixAwa [41], M^2-DTN [42], and MSST-Net [15]. Among them, FUSE and SFIM are three traditional model-based methods, while the rest are DL-based approaches. PixAwa and M^2-DTN are HSI fusion methods for unregistered images; MoG-DCN is a model-driven interpretable DL-based approach; and hyperformer, M^2-DTN, and MSST-Net are Transformer-based methods.

1) Results on Datasets With Different Degrees of Distortion: We perform a comprehensive evaluation of the proposed and competing methods under different degrees of elastic deformation on the PaviaC, Houston, and Chikusei datasets, respectively. The visual comparison (residual maps) is shown in Figs. 7, 8, and 9. We calculate residual maps based on mse, where the darker the color, indicates the lesser residuals, and the higher the fusion quality. On three datasets, the performance of FUSE, SFIM, and hyperformer decreases gradually as the degree of distortion increases, indicating their stability is poor. By contrast, MoG-DCN and MSST-Net have lower reconstruction residuals. Compared to the other algorithms discussed above, MSST-Net achieves superior fusion results due to its integration of multiscale band embeddings and a cross-modality interaction mechanism. However, it is still

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:13:02 UTC from IEEE Xplore. Restrictions apply.

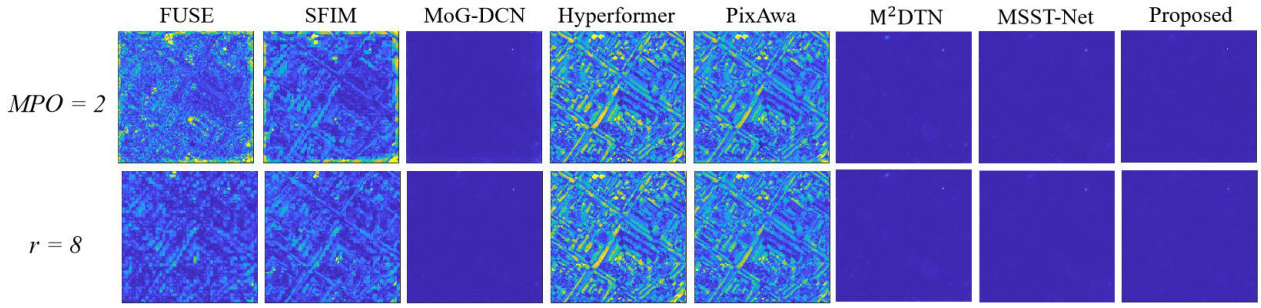


Fig. 10. Visual comparison of results of different methods on Pavia Centre with varying degrees of elastic distortion.

TABLE II
QUANTITATIVE COMPARISON ON THE PAVIA CENTRE DATASET WITH HOMOGRAPHY DEFORMATION AND TURBULENCE DEFORMATION (COLOR CONVENTION: **BEST** AND SECOND-BEST)

Degrees of distortion	Index	Traditional methods			DL-based methods					
		GLP	FUSE	SFIM	MoG-DCN	Hyperformer	pixAwa	M²DTN	MSST-Net	Ours
well-registration	PSNR	29.2897	28.8160	28.4004	42.2127	35.2354	30.4081	41.535	42.8875	**43.6771**
	CC	0.9214	0.9155	0.9039	0.9888	0.9811	0.9390	0.9886	0.9891	**0.9920**
	SAM	5.8047	7.9479	5.2229	3.0291	7.3114	8.2797	3.1978	2.9451	**2.6607**
	RMSE	0.0344	0.0363	0.0381	0.0079	0.0168	0.0302	0.0085	0.0073	**0.0066**
	ERGAS	5.3787	5.5733	5.9623	1.8147	2.9603	4.8526	1.8324	1.7307	**1.5100**
HOMOGRAPHY										
$MPO = 2$	PSNR	24.1124	25.2724	24.6019	42.3684	35.0476	29.8024	41.6462	42.0106	**43.3698**
	CC	0.6901	0.8034	0.7524	0.9893	0.9805	0.9378	0.9886	0.9879	**0.9918**
	SAM	7.8896	10.5774	7.6549	3.0629	7.2991	9.9353	3.2449	3.2762	**2.7574**
	RMSE	0.0635	0.0556	0.0591	0.0087	0.0178	0.0324	0.0083	0.0080	**0.0068**
	ERGAS	10.2535	9.0466	9.2637	1.7539	2.9963	4.9899	1.9012	1.8319	**1.5174**
TURBULANCE										
$r = 8$	PSNR	24.6662	22.4584	23.9780	41.7381	35.0755	28.6717	41.4914	42.1533	**43.3210**
	CC	0.7602	0.7408	0.6913	0.9890	0.9805	0.9084	0.9885	0.9882	**0.9916**
	SAM	7.7500	17.2983	7.5593	3.1916	7.7516	10.1417	3.2626	3.2279	**2.7652**
	RMSE	0.0587	0.0791	0.0651	0.0083	0.0177	0.0372	0.0085	0.0079	**0.0069**
	ERGAS	9.1262	14.0754	10.3909	1.8034	3.0426	5.8834	1.9209	1.8094	**1.5572**

not comparable to the method proposed in this article. The objective evaluations are shown in Table I, where the best indicators are marked with boldface, and the second-highest metrics are underlined. The results for each dataset are similar to those for comparing visual quality. The objective results show that the fusion effect of PixAwa and hyperformer is mostly worse than that of MSST-Net, and cannot achieve satisfactory fusion results. Overall, the PRFCoAM proposed in this article demonstrates strong performance in handling various degrees of distortion on different datasets.

2) Results on Datasets With Different Types of Distortion: To verify the performance and robustness of the proposed PRFCoAM against varying types of deformation, we performed experiments with PRFCoAM and competing methods on the Pavia dataset. Fig. 10 shows the results of the visual comparison under MPO $= 2$ for the homologous deformation and under $r = 8$ for the turbulent deformation. Table II shows the objective results of different methods for different types of deformation. In comparison to other methods, the fusion results of FUSE, SFIM, hyperformer, and pixel exhibit more obvious residual errors, particularly in their ability to withstand homography distortion. MoG-DCN achieves relatively stable results due to its interpretable architecture, enabling strong generalization despite not being specifically designed for registration fusion. The proposed achieved improvements compared to MoG-DCN of 3.07%, 0.26%, 11.70%, 19.41%, and 13.57% in average PSNR, CC, SAM, RMSE, and ERGAS on the three datasets, respectively. The other results of quantitative comparisons are also basically consistent with the visualization

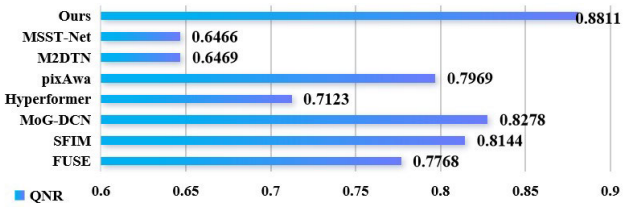


Fig. 11. Quantitative comparison of results of different methods on the real dataset.

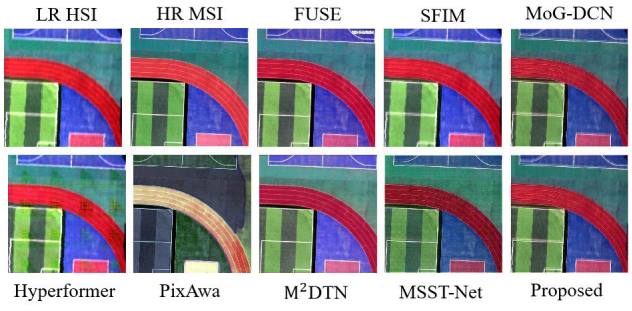


Fig. 12. Visual comparison of results of different methods on the real dataset.

results. In general, compared with other methods, PRFCoAM proposed in this article can achieve the most excellent performance when homography and turbulence deformation occur, and can achieve satisfactory fusion results.

3) Results on the Real-World Dataset: The subjective and objective metrics of the different methods on the real-world dataset are presented in Figs. 11 and 12. Compared to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:13:02 UTC from IEEE Xplore. Restrictions apply.

TABLE III

QUANTITATIVE COMPARISON OF THE PAVIA CENTRE DATASETS UNDER DIFFERENT DOWNSAMPLING RATIOS (COLOR CONVENTION: **BEST** AND SECOND-BEST)

Downsampling ratios	Index	Traditional methods		DL-based methods					
		FUSE	SFIM	MoG-DCN	Hyperformer	pixAwa	M^{2}DTN	MSST-Net	Ours
×2	PSNR	28.4428	28.5043	43.2088	39.1450	32.3294	43.0524	39.5677	**43.4429**
	CC	0.9052	0.9034	0.9888	0.9798	0.9618	0.9902	0.9766	**0.9908**
	SAM	5.9100	4.8306	3.0364	3.1005	77.0391	2.8609	3.2097	**2.7685**
	RMSE	0.0380	0.0378	0.0079	0.0108	0.0243	0.0071	0.0106	**0.0068**
	ERGAS	11.7750	12.0551	1.8234	1.7328	3.9865	1.6378	1.7064	**1.5863**
×4	PSNR	23.6112	23.3650	41.6752	35.6738	26.4920	41.1655	41.8897	**43.2514**
	CC	0.7418	0.6709	0.9886	0.9817	0.8511	0.9879	0.9883	**0.9912**
	SAM	13.9250	8.3822	3.2323	6.0510	13.0642	3.3213	3.2724	**2.8062**
	RMSE	0.0705	0.0809	0.0084	2.8477	0.0478	0.0088	0.0082	**0.0070**
	ERGAS	11.1187	11.1842	1.8276	0.0161	7.6707	1.9171	1.8363	**1.5611**
×8	PSNR	24.6522	25.0222	41.5820	38.6018	25.9868	42.9532	39.413	**43.1765**
	CC	0.8182	0.7404	0.9881	0.9758	0.8716	0.9902	0.9760	**0.9909**
	SAM	15.8733	8.0306	3.2075	3.4776	13.7479	2.8792	3.2708	**2.7899**
	RMSE	0.0991	0.0579	0.0085	0.0115	0.0504	0.0072	0.0108	**0.0070**
	ERGAS	7.9199	4.5436	1.9206	1.8489	7.9625	1.6413	1.7373	**1.5819**

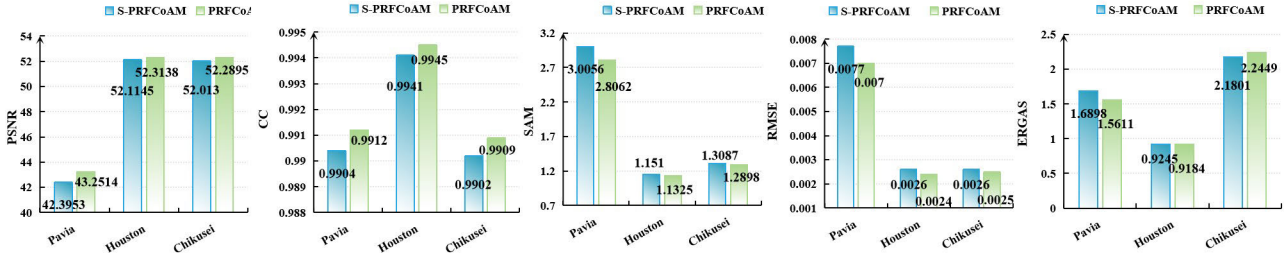


Fig. 13. Fusion results of PRFCoAM and its variant with single-stage iteration (S-PRFCoAM) on the Pavia Centre, Houston, and Chikusei datasets.

TABLE IV

SPACE COMPLEXITY AND TIME COMPLEXITY OF DIFFERENT METHODS ON THE PAVIA DATASET

Method	FUSE	SFIM	MoG-DCN	Hyperformer	pixAwa	M^{2}DTN	MSST-Net	Ours
Train (s)	/	/	35.12s	6.53	20.21	35.31	15.87	45.75
Test (s)	0.15	0.15	0.061	0.066	0.005	0.053	0.047	0.072
Flops (G)	/	/	585.68	143.48	79.74	122.00	930	138.58
Params (M)	/	/	10.39	55.44	14.04	17.30	82	7.98

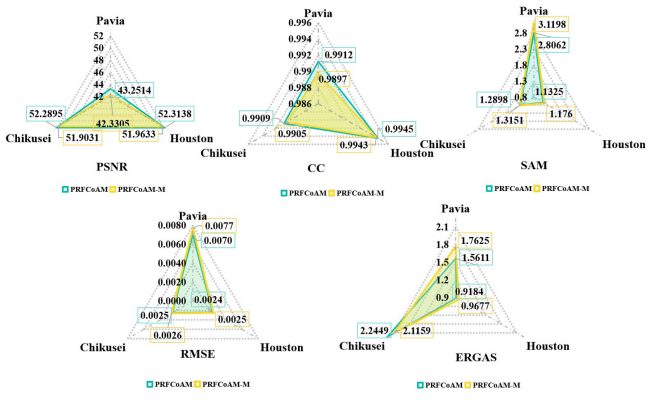


Fig. 14. Fusion results of PRFCoAM and its variant that removes MULAR (PRFCoAM-M) on the Pavia Centre, Houston, and Chikusei datasets.

the other methods, the output image of PRFCoAM on the XijingU dataset is clearer, with color and brightness closer to reality. As illustrated by the evaluation metrics in Fig. 11, other methods demonstrate strong fusion performance. However, a noticeable gap remains compared with the proposed PRFCoAM. The experimental results indicate that the proposed method achieves superior performance on real datasets compared to other approaches, reaching state-of-the-art results.

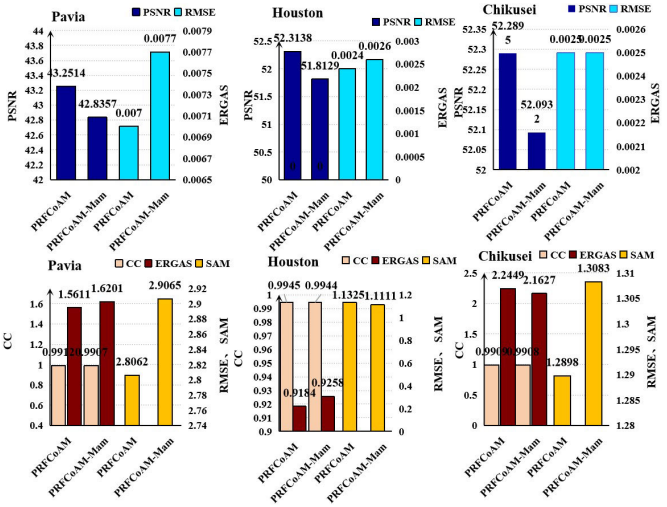


Fig. 15. Fusion results of PRFCoAM and its variant that replace SCSFE and SBSFE with Mamba (PRFCoAM-Mam) on the Pavia Centre, Houston, and Chikusei datasets.

4) Results on Different Downsampling Ratios: To validate the robustness of the proposed method under different spatial downsampling ratios, we conducted systematic comparative experiments at the downsampling ratios of ×2, ×4, and ×8. As shown in Table III, our method maintains superior comprehensive performance at the downsampling ratios of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:13:02 UTC from IEEE Xplore. Restrictions apply.

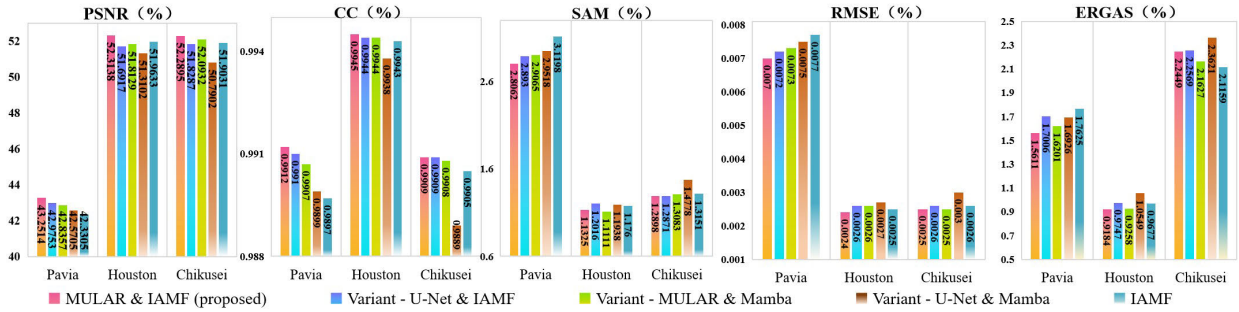


Fig. 16. Integration of MAPRF and IAMF (PRFCoAM) and the variants that replace MAPRF and IAMF, respectively, on the Pavia Centre, Houston, and Chikusei datasets.

$\times 2$, $\times 4$, and $\times 8$. These experimental results conclusively demonstrate that the proposed method can effectively adapt to varying downsampling ratios and exhibit significantly better adaptability and robustness than existing methods.

5) Time Comparison: As shown in Table IV, we train all models until complete convergence to the optimum on the Pavia dataset, using parameters, Flops, and inference time to evaluate space and time complexity. Overall, the proposed PMI-RFCoNet effectively balances performance and model computation. This method achieves a balance between performance and computation time and obtains superior fusion quality with an acceptable computational complexity.

C. Ablation Studies

In this section, several ablation studies on PRFCoAM will be conducted to validate the effectiveness of each module, utilizing three datasets containing elastic distortions with $\alpha = 1000$.

1) Progressively Registration-Fusion Co-Optimization: In the proposed PRFCoAM, we perform progressively registration-fusion co-optimization at an extensive to detailed level. To verify the effectiveness of the progressive co-optimization, we conducted an ablation study that used the single-stage fusion structure as a variant of S-PRFoAM and named it single-PRFCoAM (S-PRFCoAM). For S-PRFCoAM, we do not perform progressive registration-fusion co-optimization but perform the registration and fusion at only one level. As shown in Fig. 13, PRFCoAM performs generally superior to S-PRFCoAM on all datasets. Compared to S-PRFCoAM achieved improvements of 0.83%, 0.03%, 3.44%, 7.03%, and 1.40% in average PSNR, CC, SAM, RMSE, and ERGAS on the three datasets, respectively. This is because the progressive registration-fusion can adaptively correct the deformation from an extensive to a detailed level, refine the results of fusion at each level, enhance the overall understanding of the image, and thus prove its effectiveness.

2) MULAR Block: In the proposed PRFCoAM, we use the deformation field learned by the MULAR for spatial correction of unregistered images. To verify the effectiveness of the MULAR, we compare the performance of PRFCoAM with its variant (denoted as PRFCoAM-M), which removes the MULAR, and cascades the extracted features directly into the IAMF block. The experimental results on the three datasets are presented in Fig. 14. Overall, PRFCoAM outperforms PRFCoAM-M on all datasets, demonstrating that MULAR can achieve more accurate registration of input images, thus obtaining better results.

3) SCSFE and SBSFE: To prove the effectiveness of SCSFE and SBSFE for fusion, we compare the performance of PRFCoAM with its variant (denoted as PRFCoAM-Mam), which replaced SCSFE and SBSFE with the original Mamba, respectively. As illustrated in Fig. 15, the overall performance of PRFCoAM-Mam is inferior to that of PRFCoAM. The observed differences in effectiveness indicate the necessity of using SCSFE and SBSFE in the experiment. This is because the spatial attention and spectral attention modules can guide SCSFE and SBSFE to grasp the most critical spatial and spectral features and remove useless information. In addition, the spatial information of the input image has continuity, and there is a strong correlation between spectral bands, so the unidirectional information induction attribute of Mamba is not enough to analyze HSIs and MSIs, leading to limited spatial and spectral utilization. However, the SCSFE can fully consider the spatial context information by scanning four directions of the input MSI, and the bidirectional scanning of spectral features by SBSFE can fully extract the spectral information.

4) Co-Optimization of MULAR and IAMF: By gradually iterating MAPRF and IAMF, PRFCoAM adaptively corrects the deformation from an extensive to a detailed level and refines the results of fusion at each level to achieve progressive registration-fusion co-optimization. To verify the effect of co-optimization in the MAPRF and IAMF modules, we designed an ablation experiment. We replaced the original MAPRF and IAMF modules with the U-Net and the original Mamba module, respectively, and constructed five sets of comparative experiments (MAPRF and IAMF; Variant-U-Net and IAMF; Variant-MAPRF and Mamba; Variant-U-Net and Mamba; and only IAMF). Carrying out the fusion and registration tasks independently means that the results of fusion rely on the accuracy of the previous registration, further leading to the accumulation of errors.

Our experimental results validate this theoretical limitation as illustrated in Fig. 16. Introducing either MULAR or IAMF individually results in limited improvement in system performance, while introducing both MAPRF and IAMF simultaneously fully realizes their synergistic optimization effects, leading to a significant enhancement

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:13:02 UTC from IEEE Xplore. Restrictions apply.

and achieving optimal overall performance. The effect of co-optimization arises because improved registration accuracy provides higher-quality inputs for fusion, while the enhanced fusion output reciprocally refines registration. Such bidirectional optimization ensures more reliable and accurate final results.

IV. Conclusion

In this article, we propose a PRFCoAM which is composed of the MAPRF module. The proposed MAPRF integrates the MULAR block and the IAMF block. By performing MAPRF at each level, the network can adaptively correct the deformation of each layer and fine-tune the fusion results of each layer to achieve progressive registration-fusion co-optimization. Specifically, MULAR adaptively learns spectral and spatial degradation functions and embeds the local attention to focus on the correlations and differences between corresponding regions, thus correcting nonlinear deformation from extensive to detailed. IAMF further scans the spatial and spectral global features of the well-registration results for efficient fusion at each level. Extensive experimental results show the effectiveness and generalization of PRFCoAM. It can resist the effects of varying degrees of registration error to varying degrees and generate high-quality HR HSIs.

References

[1] S. Mei, X. Li, X. Liu, H. Cai, and Q. Du, "Hyperspectral image classification using attention-based bidirectional long short-term memory network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5509612.

[2] Y. Yang, J. Qu, W. Dong, T. Zhang, S. Xiao, and Y. Li, "TMCFN: Text-supervised multidimensional contrastive fusion network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511015.

[3] S. Mei, Y. Geng, J. Hou, and Q. Du, "Learning hyperspectral images from RGB images via a coarse-to-fine CNN," *Sci. China Inf. Sci.*, vol. 65, no. 5, pp. 1–14, May 2022.

[4] Y. Huang et al., "Two-branch attention adversarial domain adaptation network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540813.

[5] J. Peng, Y. Huang, W. Sun, N. Chen, Y. Ning, and Q. Du, "Domain adaptation in remote sensing image classification: A survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9842–9859, 2022.

[6] J. Qu, W. Dong, Q. Du, Y. Yang, Y. Xu, and Y. Li, "Cyclic consistency constrained multiview graph matching network for unsupervised heterogeneous change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5605315.

[7] Y. Dong, W. Shi, B. Du, X. Hu, and L. Zhang, "Asymmetric weighted logistic metric learning for hyperspectral target detection," *IEEE Trans. Cybern.*, vol. 52, no. 10, pp. 11093–11106, Oct. 2022.

[8] X. Fu, T. Zhang, J. Cheng, and S. Jia, "MMR-HAD: Multi-scale mamba reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, early access, Feb. 13, 2025, doi: 10.1109/TGRS.2025.3541353.

[9] R. Ran, L.-J. Deng, T.-X. Jiang, J.-F. Hu, J. Chanussot, and G. Vivone, "GuidedNet: A general CNN fusion framework via high-resolution guidance for hyperspectral image super-resolution," *IEEE Trans. Cybern.*, vol. 53, no. 7, pp. 4148–4161, Jul. 2023.

[10] W. Dong, S. Liu, S. Xiao, J. Qu, and Y. Li, "ISPDiff: Interpretable scale-propelled diffusion model for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5519614.

[11] B. Aiazzi, S. Baronti, and M. Selva, "Improving component substitution pansharpening through multivariate regression of MS +Pan data," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3230–3239, Oct. 2007.

[12] L. Loncan et al., "Hyperspectral pansharpening: A review," *IEEE Geosci. Remote Sens. Mag. Replaces Newsletter*, vol. 3, no. 3, pp. 27–46, Sep. 2015.

[13] N. Akhtar, F. Shafait, and A. Mian, "Bayesian sparse representation for hyperspectral image super resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3631–3640.

[14] S. Mei et al., "Lightweight multiresolution feature fusion network for spectral super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501414.

[15] S. Jia, Z. Min, and X. Fu, "Multiscale spatial–spectral transformer network for hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 96, pp. 117–129, Aug. 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1566253523000921

[16] H. Wu, S. Wu, K. Zhang, X. Liu, S. Shi, and C. Bian, "Unsupervised blind spectral–spatial cross-super-resolution network for HSI and MSI fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511914.

[17] X. Fu, H. Liang, and S. Jia, "Mixed noise-oriented hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5526916.

[18] S. Liu, S. Miao, S. Liu, B. Li, W. Hu, and Y. Zhang, "Circle-net: An unsupervised lightweight-attention cyclic network for hyperspectral and multispectral image fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 4499–4515, 2023.

[19] X. Wang, X. Wang, R. Song, X. Zhao, and K. Zhao, "MCT-net: Multi-hierarchical cross transformer for hyperspectral and multispectral image fusion," *Knowledge-Based Syst.*, vol. 264, Mar. 2023, Art. no. 110362.

[20] J. Qu, X. Wu, W. Dong, J. Cui, and Y. Li, "IR&ArF: Toward deep interpretable arbitrary resolution fusion of unregistered hyperspectral and multispectral images," *IEEE Trans. Image Process.*, vol. 34, pp. 1934–1949, 2025.

[21] J. Ying, H.-L. Shen, and S.-Y. Cao, "Unaligned hyperspectral image fusion via registration and interpolation modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511114.

[22] Y. Zhang, L. Zhang, R. Song, C. Huang, and Q. Tong, "Considering nonoverlapped bands construction: A general dictionary learning framework for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505215.

[23] K. Zheng, L. Gao, D. Hong, B. Zhang, and J. Chanussot, "Nonregsrnet: A nonrigid registration hyperspectral super-resolution network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2021.

[24] Y. Guo, X. Fu, M. Xu, and S. Jia, "Stereo cross-attention network for unregistered hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5523515.

[25] Y. Tay, M. Dehghani, D. Bahri, and D. Metzler, "Efficient transformers: A survey," *ACM Comput. Surv.*, vol. 55, no. 6, pp. 1–28, 2022.

[26] X. Dong et al., "CSWin transformer: A general vision transformer backbone with cross-shaped windows," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 12124–12134.

[27] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[28] K. Chen, B. Chen, C. Liu, W. Li, Z. Zou, and Z. Shi, "RSMamba: Remote sensing image classification with state space model," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[29] H. Zhang et al., "A survey on visual mamba," *Appl. Sci.*, vol. 14, no. 13, p. 5683, Jun. 2024.

[30] Y. Liu et al., "VMamba: Visual state space model," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2024, pp. 103031–103063.

[31] Z. Wan et al., "Sigma: Siamese mamba network for multi-modal semantic segmentation," 2024, *arXiv:2404.04256*.

[32] J. Yao, D. Hong, C. Li, and J. Chanussot, "SpectralMamba: Efficient mamba for hyperspectral image classification," 2024, *arXiv:2404.08489*.

[33] P. Y. Simard, D. Steinkraus, and J. C. Platt, "Best practices for convolutional neural networks applied to visual document analysis," in *Proc. 7th Int. Conf. Document Anal. Recognit.*, vol. 1, 2003, pp. 958–963.

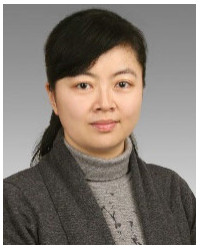
[34] T. Nguyen, S. W. Chen, S. S. Shivakumar, C. J. Taylor, and V. Kumar, "Unsupervised deep homography: A fast and robust homography estimation model," *IEEE Robot. Autom. Lett.*, vol. 3, no. 3, pp. 2346–2353, Jul. 2018.

[35] N. Chimitt and S. H. Chan, "Simulating anisoplanatic turbulence by sampling correlated Zernike coefficients," in *Proc. IEEE Int. Conf. Comput. Photography (ICCP)*, Apr. 2020, pp. 1–12.

[36] L. Alparone, B. Aiazzi, S. Baronti, A. Garzelli, F. Nencini, and M. Selva, "Multispectral and panchromatic data fusion assessment without reference," *Photogramm. Eng. Remote Sens.*, vol. 74, no. 2, pp. 193–200, Feb. 2008.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:13:02 UTC from IEEE Xplore. Restrictions apply.

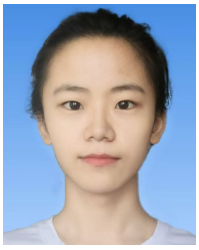
[37] Q. Wei, N. Dobigeon, and J.-Y. Tourneret, "Fast fusion of multi-band images based on solving a Sylvester equation," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4109–4121, Nov. 2015.
[38] J. G. Liu, "Smoothing filter-based intensity modulation: A spectral preserve image fusion technique for improving spatial details," *Int. J. Remote Sens.*, vol. 21, no. 18, pp. 3461–3472, Jan. 2000.
[39] W. Dong, C. Zhou, F. Wu, J. Wu, G. Shi, and X. Li, "Model-guided deep hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 5754–5768, 2021.
[40] W. G. C. Bandara and V. M. Patel, "HyperTransformer: A textural and spectral feature fusion transformer for pansharpening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, May 2022, pp. 1767–1777.
[41] W. Wei, J. Nie, L. Zhang, and Y. Zhang, "Unsupervised recurrent hyperspectral imagery super-resolution using pixel-aware refinement," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2020.
[42] W. Dong, Y. Xu, J. Qu, and S. Hou, "Learning multi-modal cross-scale deformable transformer network for unregistered hyperspectral image super-resolution," in *Proc. AAAI Conf. Artif. Intell.*, Mar. 2024, vol. 38, no. 2, pp. 1573–1581.



Zan Li (Fellow, IEEE) received the B.S. degree in communications engineering and the M.S. and Ph.D. degrees in communication and information systems from Xidian University, Xi'an, China, in 1998, 2001, and 2006, respectively.

She is currently a Professor with the State Key Laboratory of Integrated Services Networks, School of Telecommunications Engineering, Xidian University. Her research interests include wireless communications and signal processing, such as covert communication, spectrum sensing, and cooperative communications.

Dr. Li serves as an Associate Editor for IEEE TRANSACTIONS ON COGNITIVE COMMUNICATIONS AND NETWORKING and an Associate Editor-in-Chief for *China Communications*.



Yue Wen received the B.E. degree in electronic information engineering from Shanxi University, Taiyuan, Shanxi, China, in 2023. She is currently pursuing the M.S. degree with the Image Coding and Processing Center, State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an, China.

Her research interests include hyperspectral image fusion and deep learning.



Song Xiao (Member, IEEE) received the M.S. degree in communication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 2001 and 2004, respectively.

From 2006 to 2007, she was with the Viterbi School of Engineering, University of Southern California, Los Angeles, CA, USA, as a Post-Doctoral Researcher. She is now a Professor at Beijing Electronic Science and Technology Institute, Beijing 100070, China, and also a Professor at the School of Telecommunications Engineering, Xidian University.



Jiahui Qu (Member, IEEE) received the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2020.

She was an exchange Ph.D. Student of Mississippi State University, Starkville, MS, USA, from 2018 to 2019. She is currently an Associate Professor with the State Key Laboratory of Integrated Services Networks, Xidian University. Her research interests include hyperspectral image detection, image fusion, neural networks, and deep learning.



Nan Li received the Ph.D. degree in ecology from Nanjing Forestry University, Nanjing, China, in 2020.

She was an exchange Ph.D. Student of Michigan State University, Starkville, MS, USA, from 2016 to 2018. She is the Deputy Director of Anhui Province Key Laboratory of Physical Geographic Environment, Chuzhou University, Chuzhou, China. She has published over ten articles in academic journals. Her research interests include ecological environment remote sensing monitoring and vegetation remote sensing monitoring.



Wenqian Dong (Member, IEEE) received the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2020.

She has been a Visiting Scholar at Simon Fraser University, Burnaby, BC, Canada, from 2018 to 2019. Currently, she is an Associate Professor with the State Key Laboratory of Integrated Services Networks, Xidian University. Her research interests include compressed sensing, machine learning, and hyperspectral remote sensing image processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:13:02 UTC from IEEE Xplore. Restrictions apply.